

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Designed and built the back-end architecture of a multi-tenant NHS eTriage system (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** via REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack platform with clean data-management schemas (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Built and integrated back-end components with integration tests and reliability controls, AWS KMS encryption, auth hardening, and rate limiting, remediating **20+ VAPT findings**.
- Served as primary technical POC, mentored a second developer, and piloted a COO-sponsored GitHub Copilot experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Created, integrated, and tested **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with object-oriented service design, input validation, and clear, backward-compatible contracts used in production.
- Migrated core repositories from JavaScript to TypeScript and shipped a production feature running a **4-bit quantized Llama-7B** model via llama.cpp with async queued generation.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with a multi-branch data model, independently end to end; live **3+ years**.

Projects

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Built a real-time on-device system integrating camera and microphone **hardware (device I/O)** with a FastAPI back end, processing live audio and video **signals** through MediaPipe hand tracking and whisper.cpp with async queues that keep the pipeline responsive.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI back-end service with well-defined **API endpoints** and object-oriented pipeline components (typed retrieval planning, extraction, output verification, provider routing across OpenAI/Groq); hardened with **150+ automated tests** and deployed as a live cloud service.

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Designed **data-management** and ETL strategies over a **~1.4B-row** dataset with broadcast joins, skew-join handling, partitioning, column pruning, and a 6-rule data-quality filter, cutting end-to-end runtime **~25%** and the tail-latency ratio from **2.14x to 1.74x**, profiled via Spark UI.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages: C++, Java, Python, JavaScript, TypeScript, SQL, Bash

Backend & APIs: REST APIs, API design, object-oriented design, FastAPI, Node.js, integration testing

Data & Systems: MongoDB, PostgreSQL, SQLite, data pipelines, Spark, Linux

Signals & Devices: real-time audio/video processing, device I/O, MediaPipe, whisper.cpp, async streaming

Cloud & Tools: AWS, GCP, Azure, Docker, Git, GitHub Actions

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.